

1. EXCELLENCE

1.1 QUALITY AND PERTINENCE OF THE PROJECT'S RESEARCH AND INNOVATION OBJECTIVES

"Alice: If I don't find my book about «Intentions in Communication» right now, I might just blow up.

Bob: Don't worry, the MSCA Fellowship application deadline has been postponed by one month."

Introduction: Understanding *why* a specific utterance was said by a participant in a conversation is fundamental for successful communication to take place. To make this inference, the listener needs to reason about the *context* in which the utterance was said. In the above dialogue, Bob, knowing that Alice was sending an MSCA Fellowship proposal, could understand the whole meaning of Alice's sentence and provide her with relevant information. Notwithstanding the central role of **understanding intentions** in communication, this and other forms of *pragmatic* reasoning remain beyond the current state-of-the-art in Natural Language Processing (NLP). For this reason, critical applications of these technologies, such as personal assistants, remain limited to few hand-coded use-cases such as adding events to a calendar and fail to serve as general-purpose communicative assistants.

This proposal aims at making significant progress towards genuine Natural Language Understanding (NLU) by introducing a system that can consider the context in which a communicative act happened and reason to explain its observations. For this, it must **efficiently search the space of possible concepts** to discover *useful* explanations. The system must rely on an appropriate representation language for describing concepts, also known in Cognitive Science as a "Language of Thought"¹ (LOT). In this proposal, I will rely on Combinatory Logic, a Turing-complete language argued to be a good model for how humans represent concepts in their minds². Furthermore, separating helpful from unhelpful concepts requires some feedback signal. Following recent roadmaps in the field³, I argue that the system must learn from a reward signal denoting communicative success. However, learning to find useful representations in such an infinitely complex space solely from a sparse reward signal is currently unfeasible for state-of-the-art methods. The biggest roadblock is the open problem of **exploration**, namely, the question of deciding which candidate hypothesis might be the most promising to try next⁴. For instance, before having found the correct explanation for Alice's utterance, Bob might have tried other plausible alternatives, such as "Are you teaching tomorrow?" or "Do you need to return it?". However, it would have been wasteful of their time to figure out whether Alice liked to do gardening or whether her dog was hungry. To solve this issue, the proposed system will **learn its own exploration strategies** in the same way it will learn other concepts. Therefore, there will be no fundamental distinction between its concepts about the external environment and the learning mechanisms. In this way, every part of the system is learnable and subject to continuous adaptation.

To achieve all these goals, I will design a system inspired by **evolution** where useful concepts, including the algorithms searching for new concepts, will reproduce more frequently than unhelpful ones. The system will rely on *algorithmic mutations* to search for new concepts, allowing it to *compose* previously known concepts, thus maximizing the probability of finding new useful ones. To design such a system, I will draw on and extend my previous research on the emergence of evolutionary units in a complex system⁵. This project bridges Linguistics and Complex Systems with connections to Cognitive Science and Evolutionary Theory. To maximize its success probability, I will conduct it at the **COLT Lab in Universitat Pompeu Fabra (UPF)**, a leading group in Computational Linguistics, with a secondment at **Santa Fe Institute (SFI)**, a global leader in Complexity Science.

State of the Art: Early NLP systems relied on a collection of *hand-written* rules in a computational LOT that allowed the system to map natural language text into a formal representation that the computer could operate with⁶. However, the coverage of the natural language that the computer could understand remained very limited because a human had to manually enter both the lexical database containing the meaning of words and the grammar that

combined the words into expressions. Recent work² argues that a convenient LOT is Combinatory Logic, a variable-less compositional representational system that can describe any computable function (Turing-completeness). This framework represents an advancement to early practice because concepts can be *learned* from evidence. Given some observations about concepts (e.g., spring follows winter), it assigns them representations such that after relating them through function application, the results are consistent with observations. For instance, the concepts of spring, winter, and succ must be compatible with " $(succ\ winter) = spring$ " (see Figure

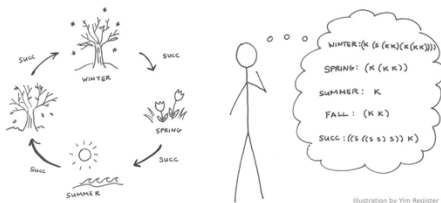


Illustration by Yin Register

Figure 1 Conceptual representation in Combinatory Logic (reproduced from [2]).

1). However, finding such representations requires enumerating the entire space of representations, which cannot scale to large repertoires of interacting concepts due to the sheer size of the search space. Furthermore, it is unclear how such a process could be applied to learning from a reward signal rather than direct evidence. Instead, Deep Learning, based on neural network systems, is the dominant paradigm nowadays. In these models, the representations associated with words and utterances emerge from the training dynamics of the system, which often consist of an optimization process on its parameters aimed at maximizing the likelihood of observed data. Recently, very large instances of neural language models (NLMs) trained to predict the continuation of text extracted from large text

corpora have revolutionized the field with their ability to write human-like continuations when prompted with natural language utterances⁷. Yet, despite their impressive capabilities at tackling many tasks, it has been repeatedly argued that such models cannot genuinely account for NLU because they are oblivious to the context surrounding the input utterance^{8,9}. Consequently, they cannot draw pragmatic inferences by relating the literal meaning of input utterances with the external context, thus making it unlikely that an NLM could have generated Bob's response above.

A promising way forward comes from the Reinforcement Learning (RL) field, which studies learning from a reward signal. Agents in RL can have access to their context by observing the current state of the world or a partial observation of it, which they use to pick the following action. Thus, this formalism can theoretically allow for modeling pragmatic reasoning in artificial agents if agents are rewarded by their ability to communicate successfully and have sufficient capacity to infer how the meaning of input utterances is affected by the current context. However, existing RL algorithms are notably *sample-inefficient*, especially compared to humans¹⁰, meaning that they require a vast number of observations to learn. For this reason, complex learning problems, such as learning communication from a reward signal, are beyond the state-of-the-art. To improve sample efficiency, we must tackle the open problem of *exploration*, which entails identifying alternative actions that can lead to higher rewards. A system that explores fewer but more promising alternatives can learn more efficiently. While previous studies have investigated exploration in RL extensively, this remains a challenging unsolved problem due to uncertainty, sparsity of rewards, and the compounded complexity of multiple acting agents⁴.

Recently, the **Hanabi challenge**¹¹ has been introduced to test pragmatic reasoning in RL agents. This challenge provides a simplified testbed that does away with the complexities of linguistic communication. In this challenge, agents play a card game where the goal is to build sorted piles of same-color cards. At each turn, players



Figure 2 The Hanabi Challenge

should either add the next card in sequence to some of the piles or discard if no valid play is possible. Crucially, each agent knows other agents' cards, but they do not know their own. To successfully solve the game, they can provide hints to other players about the cards they hold, but the number of hints is limited and can only be about a single feature. Thus, to maximize information transfer, the agents must understand *why* another player gave a specific hint. For instance, in Figure 2, P1 cannot tell P2 that he holds two "1s" because the yellow one has already been played. Instead, she can inform him about the red card, expecting that P2 will pragmatically infer that this is a red 1. The state-of-the-art method¹² is still far from achieving the same performance as hand-coded bots, showing that pragmatic reasoning is far from solved: On one hand, we cannot expect hand-coding strategies to scale to more complex tasks. On the other hand, we cannot expect state-of-the-art RL methods to apply to natural language communication if they fail in these simplified environments.

Quality and pertinence of the R&I objectives: In this project, I hypothesize that **pragmatic reasoning**, such as learning other agents' **intentions**, must be represented in a Turing-complete LoT such as Combinatory Logic. To explore this space of concepts as efficiently as possible, I hypothesize that **the exploration strategies must be emergent** (instead of hand-designed). In this way, they can become *more complex* and *evolve continually* as the agent acquires experience. In other words, these strategies must undergo an **open-ended evolution (OEE)** process. I will investigate how evolutionary units can emerge using Combinatory Logic as a computational basis. In a previous study⁵, I have shown sufficient conditions for the emergence of self-reproducing metabolisms. Here, I hypothesize that it is enough to scale up the existing system and allow mutations in the computation for OEE to emerge. I will then leverage this process to propel an RL algorithm with efficient exploration skills. I will test this algorithm on the Hanabi challenge to evaluate whether it can learn other agents' intentions. To succeed at this game, each player must interpret the communicative intentions of the others. Because it features complex dynamics on a minimal action and state space, it is an ideal benchmark for this system. Finally, together with my host group at UPF, I will examine the rules learned by the system to find evidence of intention comprehension. More precisely, my research objectives are:

- RO1** Investigate the conditions for the emergence of an open-ended evolutionary process.
- RO2** Design and implement an RL system based on open-ended evolution.
- RO3** Develop a learning system that can learn to play Hanabi.
- RO4** Demonstrate that the system has learned heuristics based on interpreting other agents' intentions.

This program sits at a cutting edge to current Artificial Intelligence efforts, as follows. First, as argued above, the state-of-the-art at understanding other agents' intentions is unsatisfactory. Solving this problem is essential in building *human-centric AI* (cf. European Commission's Communication COM.2019.168) because a) it pursues systems capable of efficiently adapting to the human user's intentions; b) the proposed method relies on compositional representations obeying systematic rules, which are more *transparent* than artificial neural networks; and c) this proposal aims at building more *robust* systems that can continually adapt to novel and unexpected situations, in contrast to current systems trained once and for all. Furthermore, this project aims at discovering the conditions for an open-ended evolutionary process to arise, including the origin of evolutionary units. This is a fundamental open question for the Artificial Life community and an open problem for science in general.

Even though these are ambitious goals, I have a clear path to achieving them (cf. Section 1.2) and have defined concrete milestones that will signal their completion where relevant. First, because there is no commonly accepted definition for an open-ended evolutionary process, I will consider objective **RO1** achieved if I can observe that increasingly more complex forms emerge in the system as a product of its emergent exploration dynamics (**M1**). To achieve this objective, I hypothesize that it is sufficient to introduce mutations to a previously proposed system featuring emergent self-reproducing metabolisms and by scaling up the size of the simulations (cf. Section 1.2). Second, I will measure the completion of **RO2** by probing the system on the most elementary form of the RL problem, namely bandit tasks (**M2**), in which the agent must choose a single action to get an immediate reward. This simple task already shows how an agent deals with exploring the action space. Third, Hanabi is a challenging game for state-of-the-art RL systems, and thus it sets a high bar for the proposed system (**RO3**). However, to make genuine progress in systems that can meaningfully communicate with humans, this task also represents the bare minimum since all communication is non-verbal and, thus, less complex than interpreting linguistic signals. Nonetheless, we set a modest win rate of 5% at 5-player Hanabi as our target (**M3**), which is comparable to the state-of-the-art learning algorithms (and still far from >90% of hand-coded bots). With the host group, who have theoretical and experimental pragmatics experts, we will analyze whether the heuristics learned by the system align with a system capable of interpreting other players' intentions. The fact that the system will construct routines in a compositional language makes it more **amenable to interpretation** than current neural systems, facilitating this task.

1.2 SOUNDNESS OF THE PROPOSED METHODOLOGY

Overall methodology: In **WP1 Emergent Evolution**, I will seek to uncover the conditions that lead to the emergence of an open-ended evolutionary process (**RO1**). For this, I will draw on my previous work on Combinatory Chemistry⁵, an Artificial Chemistry designed to study the origin of evolutionary units in which self-reproducing metabolic structures emerge autonomously from its dynamics. In summary, the system is initialized with a multiset $\mathcal{P}$ containing an equal number of combinators S , K , and I . The elements in the system are subject to *reactions*, which are duly logged. Given x and y in $\mathcal{P}$, an *application reaction* can combine them into an expression (xy) . It is helpful to note that application is left-associative, and thus $((xy)z)$ can be written simply as (xyz) . Additionally, a *cleavage reaction* can split expression $e = (x'y')$ in $\mathcal{P}$ into its subcomponents x' and y' . Crucially, in addition to *random recombination*, an infinitely large set of so-called *reduction reactions* occur at a much faster rate. These reactions extend Combinatory Logic rewriting rules, from which the system derives its Turing-complete representational power. Let $e = (\alpha X \beta)$ be an expression in the system's state $\mathcal{P}$ where X is some subexpression and α and β are some (potentially empty) contexts, then the following reduce reactions are defined:

$$\begin{aligned} \alpha(Ix)\beta &\Rightarrow \alpha x \beta + I && \text{(I – reduction)} \\ \alpha(Kxy)\beta &\Rightarrow \alpha x \beta + y + K && \text{(K – reduction)} \\ \alpha(Sf gx)\beta + x &\Rightarrow \alpha(fx(gx))\beta + S && \text{(S – reduction)} \end{aligned}$$

Metabolisms emerge as sets of reduction reactions that take place cyclically. Let's assume, for example, the expression $(SII(SII))$ to be present in $\mathcal{P}$. Should also (SII) be present in $\mathcal{P}$, then a chain of reactions can occur, first an S -reduction that absorbs one copy of (SII) into the expression, and then two I -reductions to bring the expression to its original state (see Figure 3). This cycle represents the simplest metabolism that emerges in Combinatory Chemistry. Interestingly, similar structures that self-reproduce also emerge from these simple dynamics. However, the system does not yet display emergent evolutionary behavior. I hypothesize that two properties are needed to observe this phenomenon. The first one is *large-scale simulations*. I have empirically observed that more extensive simulations lead to more diverse structures. For instance, while self-reproducing forms do not emerge on systems starting with 10K combinators, they emerge in systems with 1M combinators. Still, even at this scale, they go extinct for lack of sufficient production of the corresponding “food expressions.” Larger systems should remediate this latter aspect, leading to predator-prey dynamics and a broader diversity of more complex structures. The current bottleneck is in the computational load of simulating the system using the Gillespie Exact Algorithm. Fortunately, this algorithm has been extensively studied in the past due to its importance in computational chemistry. In task **T1.1 Scaling up the simulation**, I will explore existing solutions to speed up the system. In particular, I will implement a solution based on leveraging graphics processing units¹³ that I will run on UPF's high-performance cluster. To reduce environmental impact risks, I will track the energy usage and experiment with other well-known combinators to improve efficiency. Nonetheless, we cannot expect evolution to appear just from simulating larger systems. An equally important point is allowing “errors” in the computation to *produce mutations*. To this end, in task **T1.2 Exploring the effects of mutations**, I will include a second kind of S -reduction that will absorb an arbitrary element y , and which will occur at a slower rate than normal reduction reactions, but at a considerably higher one than random recombination: $(\alpha(Sf gx)\beta + y \Rightarrow \alpha(fx(gy))\beta + S$. To see why it is essential to introduce this kind of reaction, let us again examine the previous expression $(SII(SII))$, but this time undergoing a mutation in which instead of absorbing a copy of $x = (SII)$ it absorbs an arbitrary

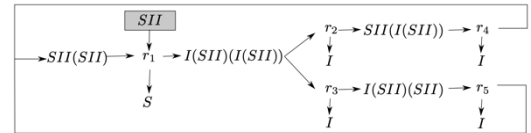


Figure 3 A simple metabolism.

expression y , leading to $(I(SII)(Iy))$, and after a chain of reduction reactions, to (yy) . In other words, this particular mutation on the $(SII(SII))$ metabolism will compute expressions where y is applied to itself for any y found in the system. Suppose the system is looking for viable metabolisms, an essential property of an evolving system. Then, this strategy provides a beneficial inductive bias to apply because all known metabolisms emerge as an expression applied to itself. Thanks to this mutation, the system will test these structures much more frequently than if it had to wait for them to be produced by random recombination, thus improving search efficiency. This emergent search process is a fundamental aspect of evolution that I will leverage for an exploration algorithm. To establish whether a system designed in this way leads to an open-ended evolutionary process (M1, cf. Sect. 3), I will use an information-theoretic metric that I have proposed before to identify emergent metabolisms. Then, I will use the history of previous reactions to track how they first emerged. I will conclude that there is evidence supporting the hypothesis of an open-ended evolutionary process if mutations over existing metabolisms lead to novel and possibly more complex metabolisms. To facilitate disseminating this research to the broader public (viz. Sect. 2.2), I will also create animated visualizations of metabolic structures of interest (T1.3 Visualization).

So far, I have described how the system may learn to search the space of possible metabolisms. In **WP2 Reservoir Learning**, I will explore how an RL algorithm can leverage this process (RO2). The central hypothesis is that some of the emergent metabolisms will encode possible solutions to real-world problems, and we can encourage their replication when the solutions prove useful. Furthermore, mutations over these metabolisms will allow the system to *explore* alternative solutions in a structured way by reusing the system’s prior knowledge. Defining the learning algorithm requires answering three questions, namely, 1) how to encode the state? 2) how to compute the model’s actions for a given state?, and 3) how to feed rewards? To evaluate the system, I will rely on bandit and contextual bandit tasks, in which rewards are fed immediately after each action (M2). In **T2.2 Encoding/decoding states and actions**, I will study how we can encode the state of the world into a Combinatory Logic expression, feed this expression as input to a subset of programs in the current population $\mathcal{P}$, and obtain the system’s preferred action. To do this, I will focus on a hypothesized family of self-reproducing expressions of the form (Γz) that self-reproduce with a payload of arbitrary information z . Then, given some state s , the model will encode the state information using an arbitrary mapping, thus obtaining e_s , and then feed this encoded state as input to z , obtaining (ze_s) . Then, the model will interpret the result of evaluating this expression (if any) as a chosen action. The final decision will be selected using an aggregation rule over the produced actions, such as majority voting. To maximize success probabilities, in **T2.1. Identification of Γ combinator**, I will seek to characterize the Γ combinator even before empirically observing it to emerge in the system. Finally, in **T2.3 Feeding rewards**, I will investigate how we can reward the system for producing desirable actions. Conveniently, the fact that the system has an analogous of “food” gives us a natural way of rewarding the system. For instance, (Γz) will need to consume copies of Γ to reproduce itself. Therefore, adding Γ combinators into the system when it acts desirably allows existing metabolisms to reproduce, thus increasing the chances that the model will take the same or similar actions in the future.

Finally, in **WP3 Playing Hanabi**, I will develop a system that learns to play the Hanabi game (RO3). In **T3.1 Communication in Hanabi**, I will investigate with the host group the dynamics of Hanabi under the lens of the non-verbal communication between the players. More specifically, we will analyze the meaning of other agent actions in the game under the lens of Gricean Pragmatics (which establishes that messages must be as informative, truthful, relevant, and unambiguous as possible). Equipped with this analysis, we will produce a bot based on hand-coded rules implemented in Combinatory Logic. This analysis will illuminate the richness of the space for non-verbal communication in Hanabi and convenient inductive priors to learn to interpret messages in this game. In **T3.2 Learn Hanabi**, I will test whether the described RL algorithm can learn other agents’ intentions using the Hanabi learning environment made available by [11]. Furthermore, together with the host team, in **T3.3 Interpretation**, we will interpret the learned rules by relating them to the priors that we identified in T3.1 to verify whether the system has meaningfully learned to play by taking into account other players’ intentions (RO4). For this, we will identify repeated or prominently used functions and give them interpretable names. While this task would be non-trivial for neural networks, here is considerably facilitated by the compositionality of representations in Combinatory Logic.

Integration of methods and disciplines to pursue the objectives: This action links the highly interdisciplinary fields of Computational Linguistics (CL) and Complex Systems (CS), crossing over to Artificial Intelligence and Cognitive Science. This proposal focuses on some of the most fundamental problems that current CL research faces, particularly when trying to account for pragmatic aspects of communication, thus potentially having a groundbreaking impact in this field. Moreover, at its heart, this project features an open-ended evolutionary process of the sort that the CS field studies in the subfield of Artificial Life. While evolution have not occupied a mainstream role in AI so far, new developments in this project should strengthen a two-way interface between CS and AI.

Gender dimension and other diversity aspects: Because of how current NLP systems are trained, they can replicate any biases observed in the training data. Instead, this proposal aims to build systems that can learn from a communicative success reward signal. I hypothesize that such systems can be discouraged from producing sexist,

stereotypical, or other kinds of harmful content by applying negative rewards, which the system will use to learn to avoid such behavior using its flexible adaptation skills.

Open Science I have a coherent history of making publicly available all the products of my research, including code, datasets, and publications, all linked in my homepage germank.github.io. I will continue ensuring open access of all publications by (1) contracting immediate open access upon publication; (2) using the UPF open access repository (UPF e-prints); and (3) uploading pre-prints to repositories such as arXiv. Furthermore, I will make software and data produced by this research freely and openly available on platforms such as GitHub with open licenses, including Creative Commons or MIT. Finally, because I strongly believe in encouraging and informed debate in society, I will further promote discussion events with the general public through planned activities to this end (cf. Section 2.2.)

1.3 QUALITY OF THE SUPERVISION, TRAINING AND OF THE TWO-WAY TRANSFER OF KNOWLEDGE BETWEEN THE RESEARCHER AND THE HOST

A detailed **Career Development Plan (PCDP)** will be designed together with my supervisor before the start of the fellowship and monitored throughout to ensure the achievement of the short- and long-term objectives. It will cover, a.o.: Research results: Publications and conferences/workshops as in Section 2.2; Research skills and techniques: Training objectives TO1.1 - TO1.4 in this section; Management and business awareness: TO2.4, TO2.5; Communication skills: TO2.3; Other professional training: UPF crash course on intellectual property and knowledge transfer, UPF workshop on external funding (TO2.2, TO2.5); Anticipated networking opportunities: Conferences, workshops (see Section 2.2), secondment at SFI, and the world-renowned researchers that regularly visit UPF.

My overall scientific training objective (TO1) is to become an expert in systems capable of understanding the intentions of other agents. To complete this ambitious project, I will deepen my knowledge in:

TO1.1 Pragmatics: The question of how we infer a speaker's intentions belongs to the broader field of Pragmatics. Pragmatics investigates how context affects the meaning of an utterance. At UPF, I will interact with many experts in the area, including, but not limited to Prof. Gemma Boleda and Prof. Louise McNally, who are the heads of the COLT and GLiF labs at UPF and Dr. Thomas Brochhagen, Dr. Laia Mayol, and Dr. Mora Maldonado who are postdocs specialized in pragmatics in those same labs. With their advice, I will analyze the pragmatic aspects of Hanabi (T3.1) to inform the development of the envisioned system from day one. In contrast to developing AI systems far from the domain experts and evaluating them *post hoc*, I will incorporate any identified desiderata into the development cycle early on, thus leading to more robust systems by design.

TO1.2 Non-verbal communication in AI: This proposal centers on non-verbal communication as a stepping stone to future work in linguistic communication. Verbal and non-verbal communication are central research topics of COLT with two related ERC projects: AMORE on referential aspects of communication and ALiEN on emergent communication between artificial agents. This project will contribute to these projects by exploring non-verbal communication in Hanabi while also benefiting from interacting with experts in communication-based AI.

TO1.3 Complex Systems: Complexity is the field that analyzes how complex behavior emerges out of the interaction of simpler components. The emergence of living, evolving, and intelligent systems is one of the central questions at the core of this field. Having already contributed from outside of the field to understand the emergence of self-reproducing metabolisms⁵, I expect to continue to significantly grow my knowledge of Complex Systems through this research project that pushes its frontiers. My secondment at SFI, the center that has been at the heart of Complex Systems research since the field was born, will provide me with crucial training. At SFI, I will be able to collaborate with my supervisor, Prof. Melanie Mitchell, an expert on the relation between AI and Complexity, and a team of experts on topics such as Evolutionary Theory (Prof. Michael Lachmann), Collective Computation (Prof. Jessica Flack), and Artificial Chemistries (postdoctoral Fellow Dr. Andres Ortiz-Muñoz).

TO1.4 Emergent learning: Current AI algorithms are designed around optimizing an objective function. In this project, I will investigate a new paradigm for understanding intelligence, where learning and exploration are *emergent* properties of a complex system instead of explicitly designed. Should this paradigm prove fruitful, I will be well-positioned to become a global leader in this topic by establishing its foundations.

My strategic training objective (TO2) is to acquire other transferable skills to become a leading, independent researcher, ready to attain a stable position in Europe and apply for funding as a PI. Thus, I will train:

TO2.1 Collaborations: I will improve my already exceptional skills in establishing long-term collaborations to produce a network that will make me an integral part of the scientific circuit.

TO2.2 Leadership: To train my supervision and leadership skills, I will engage in mentoring undergraduate (B.Sc./M.Sc.) students, organizing meetings and seminars with the goals of transferring my knowledge to my group members, and establishing collaborations with them and with other external parties.

TO2.3 Communication: I will produce higher impact publications and publish more in journals. Given the multidisciplinary aspects of my proposal, I will submit my work to high-impact interdisciplinary journals, such as PNAS. I will also engage in public dissemination activities and reach the outer public.

TO2.4 Management: I will learn for the first time to plan and manage my scientific project, project finances, establishing my network of collaborations, dissemination, and outreach activities, with the support of my supervisor,

Prof. Boleda. For financial management, I will get regular overviews of the spending on my project from the host's finance department.

TO2.5 Business Awareness: While my project is concerned with fundamental research, it could become relevant for the industry in the long term. The university's technology transfer office offers training about intellectual property that I will participate in.

Qualifications and experience of the supervisor: Prof. Gemma Boleda is the head of the Computational Linguistics and Linguistic Theory (COLT) Lab at U. Pompeu Fabra. She carries out cutting-edge, interdisciplinary research at the interface between computational linguistics and linguistics. She is the recipient of an ERC Starting Grant on computational approaches to reference, which is related to my research goals, as this project applies Machine Learning to model reference and communicative success. Her COLT team consists of three senior researchers, five PhD students, and two postdocs with a broad range of expertise from Machine Learning methods and artificial neural networks to computational semantics, pragmatics, and communicative approaches to language. She has postdoctoral supervision experience, comprising five postdocs in her ERC grant and one former MSCA Fellow. Since January 2020, she has been an ICREA Senior Researcher. Six of her publications have appeared in high-impact international journals in the JCR (Journal Citation Reports of Thomson Reuters' Web of Science) and 26 in the top 10 venues in computational linguistics according to Google Scholar (h-index 24, see <https://scholar.google.de/citations?user=NfJ9kUEAAAAJ>). She has postdoctoral experience at The University of Texas at Austin (USA) and the University of Trento (Italy), both funded by MSCA fellowships. Gemma Boleda is widely recognized as a leader in computational semantics by the international community; a.o., she was co-editor for a special issue in a top journal (*Computational Linguistics*), and she has been an area chair for ACL in Semantics. In 2015-2021, she gave 27 invited or keynote talks at international workshops. Notably, Prof. Boleda's group ensures my training in transferable skills because its researchers engage in interdisciplinary research and routinely publish in high-impact journals in and outside my field; it offers mentoring possibilities; it facilitates public engagement activities both through general programs at UPF and through the support of the university's dedicated press media agent (see Section 2.2), and it provides excellent support for grant management (see Section 3.2). Moreover, the Technology department of UPF hosts other groups with expertise in Complexity, Pragmatics, and NLU. All these groups are interconnected under the umbrella of the same overarching unit, Campus de la Comunicació of UPF, located in the Barcelona 22@ technology district, including high-tech companies, other universities, technology centers, and public administrations creating an outstanding area for RTD projects in my field.

Prof. Melanie Mitchell is the Davis Professor at the Santa Fe Institute (SFI) and is the author or editor of six books and numerous scholarly papers in Artificial Intelligence, Cognitive Science, and Complex Systems. She authored four books, 15 book chapters, and numerous articles at journals and conference proceedings. Only from 2019 to 2020 she has given 33 invited lectures. Prof. Mitchell has extensive supervision experience, counting eleven PhD students and three postdocs. She is known for her work in abstraction and analogy-making, used by humans to learn, reason, and robustly adapt their knowledge to new domains. Being an author of a popular textbook on Complexity and an expert in AI, she is the ideal supervisor for this project bridging the two areas. Prof. Mitchell will ensure that I will enjoy a tight interaction with the highly collaborative and interdisciplinary environment of the SFI, a global hub for Complexity research.

Transferring knowledge to the host organization: My previous experience at the University of Trento, Facebook AI Research (FAIR), and Naver Labs Europe (NLE) has endowed me with a broad knowledge of Machine Learning, GPU computing, management of large codebases, and good coding practices. Furthermore, I have acquired extensive knowledge of **Deep Learning methods** for NLP. These techniques are core to the current research of Prof. Boleda's group, and I will be able to foster their knowledge in several ways, including, but not limited to, research collaborations, seminars, informal meetings, and tutorials that I will organize together with Prof. Boleda and her group. Furthermore, thanks to my research experience in the industrial sector, I will provide the opportunity for Prof. Boleda's group to establish new collaborations with my former and current colleagues at FAIR and NLE, strengthening the synergy between UPF and the private sector.

1.4 QUALITY AND APPROPRIATENESS OF THE RESEARCHER'S PROFESSIONAL EXPERIENCE, COMPETENCES AND SKILLS

I am a **rigorous** and **creative** young scientist with solid **collaboration and team-working skills**. I have a proven **research record** with notable **impact** (most cited publication: 2678 citations, h-index: 13, according to <https://scholar.google.fr/citations?user=uU3rQI8AAAAJ>). Connecting the two seemingly disparate research directions of pragmatics and evolution has been my long-term research goal. On the one hand, I have published groundbreaking research on computational pragmatics at the high-impact Computational Linguistics journal¹⁴. On the other hand, I have also invented the system that this proposal builds upon. Furthermore, I have extensive experience performing high-impact research from my doctoral work and my experiences at FAIR and NLE, and solid skills in computational modeling. For all these reasons, I am uniquely qualified to run this project.

2. IMPACT

2.1 CREDIBILITY OF THE MEASURES TO ENHANCE THE CAREER PERSPECTIVES AND EMPLOYABILITY OF THE RESEARCHER AND CONTRIBUTION TO HIS/HER SKILLS DEVELOPMENT

Working in prestigious labs such as FAIR has given me a firm handle on current computational models. However, industrial research labs do not have the incentives to invest in high-risk high-impact research, instead focusing on improving already successful technologies. The MSCA fellowship will represent a massive leap in my career, enabling me to pursue an exceptionally innovative research direction that could significantly impact AI and Complex Systems research. Furthermore, it will train me on crucial skills that I will need to complete such an ambitious research program (cf. Section 1.3). Moreover, allowing me to become a **leading, independent researcher in Europe in how pragmatic reasoning can be modelled through evolution** will significantly increase my chances of obtaining a tenured or tenure-track position and funding as a PI to investigate these topics further. Furthermore, by training at the hosting intuitions in building a new bridge between Linguistics and Complex Systems, I will establish myself at the vertex of these disciplines, integrating and cross-pollinating them. Moreover, I will publish in high-impact venues and gain more visibility (cf. Section 2.2) and enrich my research network with researchers at UPF and SFI, and people connected to them. Concretely, the next step in my career will be to apply for an ERC Starting Grant, for tenure-track programs such as the Spanish *Ramón y Cajal* program, and university lecturer or professor positions.

2.2 SUITABILITY AND QUALITY OF THE MEASURES TO MAXIMIZE EXPECTED OUTCOMES AND IMPACTS, AS SET OUT IN THE DISSEMINATION AND EXPLOITATION PLAN, INCLUDING COMMUNICATION ACTIVITIES

Dissemination and exploitation of results: Per the *European Charter for Researchers*, I consider dissemination an indispensable part of research and ensuring rapid dissemination of research results a duty of a scientist. The dissemination of the project's results to the scientific community (WP6) will target **Artificial Life conferences** (e.g., ALIFE, GECCO) concurrently with WP1 to disseminate progress on open-ended evolution; **Computational Linguistics / Pragmatics** (e.g., ACL, EMNLP, and the Semantics and Pragmatics journal) concurrently with WP3 to disseminate the work done on pragmatics in the Hanabi learning environment; and **Multidisciplinary journals** (e.g., Nature, Science, PNAS on WP5): Because it is an ambitious but well-planned project (see Section 1.1), with far-reaching implications for AI, Computational Linguistics and Complex Systems, it will advance the state of the art in each of these fields. I will thus aim at publishing my results concurrently with the end of WP3 in high-impact multidisciplinary journals to present the findings of this project in a comprehensive way to a broad audience. I will also present my research through invited talks at universities and scientific events. The planned dissemination will positively impact my career, as I will actively network and become highly visible at conferences and workshops, which will foster a higher impact of the journal publications. Because I will also target high-impact multidisciplinary journals, I will become a leading researcher on pragmatic reasoning through evolutionary methods. I am aware that under Horizon 2020, I will need to ensure open access to the publications arising from the current project, and I will implement the necessary measures (described in Section 1.2).

My research program is likely to have an appeal to a broader audience. Thus, with the assistance of the UPF *Scientific Communication Observatory* (<https://www.upf.edu/web/occ/>), I will carry out activities directed at raising social awareness of the impact of my research in suitable forums: in the science and technology sections of national and, if possible, international press media, discussing with notable journalists and science divulgators on the topic of complexity science, such as Michael Garfield from the SFI Complexity Podcast or John Rennie from Quanta Magazine; in radio programmes such as *Generació Digital* (Catalunya Ràdio), which has featured interviews related to the language technology being developed at UPF; through social networks, especially the UPF Facebook and LinkedIn pages; with an activity at EscoLab 2022 (www.escolab.cat), a programme of the Barcelona Council to promote science directed at secondary school students; and in the Spanish (Barcelona) instalment of the European Researchers' Night (NIGHT) 2023 (<http://www.lanochedelosinvestigadores.es>). Beyond discussing the research outcomes in this project, I will engage the public on general debates linked to the research objectives of this project: "What is Life?" and "What would happen if we could talk to computers (and they truly understood us)?" These general and vital questions are likely to spark fruitful and far-reaching debates. All detailed outreach activities will happen during the full span of the project, as shown in Section 3.1.

Intellectual property (IP): To protect the IP of the inventions, I will seek the assistance of the *UPF Innovation Unit*, which actively pursues the protection, valorization, and exploitation of results and intellectual property generated by UPF staff. The activities carried out by this unit include: general information about IPR; assistance in developing exploitation strategies that effectively translate research results into benefits for society; evaluation of intellectual property protection mechanisms; support for market analyses and negotiations of license agreements and private R&D contracts; creation of start-up companies; guidance and support for raising external funds. The UPF policy on intellectual property rights (IPR) complies with the EU Policy and guarantees that both the research groups (for continued research) and the researchers personally can benefit from exploiting their R&D results.

2.3. THE MAGNITUDE AND IMPORTANCE OF THE PROJECT'S CONTRIBUTION TO THE EXPECTED SCIENTIFIC, SOCIETAL AND ECONOMIC IMPACTS

Expected scientific impacts: If successful, this project will have a sizeable scientific impact in at least three ways. First, the conditions that lead to the emergence of an open-ended evolutionary process are not yet known. If the hypotheses behind this project are correct (cf. Section 1.1) and the simulated system displays an emergent evolutionary process, this will have vast implications for our understanding of living systems. Furthermore, in AI, it will pave the way to new systems that improve their learning capacities because they will be emergent rather than a hand-designed system component. Consequently, this will lead to more **data-efficient RL methods** thanks to exploring more efficiently the space of possible behaviors. Finally, it will pave the way to model **pragmatic aspects of meaning**, a frontier that current methods struggle to overcome, which is required for genuine NLU in AI systems.

Expected economic/technological impacts: Because the proposed system will learn from less data, it will unlock new possibilities for adaptive agents that learn “on the fly.” In contrast to current Machine Learning models that are trained once and deployed “frozen,” preventing them from learning from new experiences, the proposed learning algorithm will be able to adapt continually. Equipped with these skills, products like personal assistants will be able to learn new behavior to satisfy the demands of the human user. Trustworthy and human-centered AI are focus areas for the EU, as detailed in topic HORIZON-CL4-2021-HUMAN-01-01 of the HORIZON-RIA action. By developing a system that can interpret human intentions, learning from fewer data points, and using a compositional backend that is more easily interpretable, this project makes significant contributions to those areas.

Expected societal impact: This project can have considerable societal implications in the context of human-machine interaction. Humans currently communicate their intentions to machines by using formal programming languages, which require them to express meaning in a great level of detail. Thus, the task of communicating with machines is delegated to specialized professionals (programmers). The long-term success of the research line brought forward by this project would allow humans to use their natural language to program computers because the machines will resolve ambiguities in the speaker's intention. After taking appropriate action to cover as much as possible the language of every speaker on the planet, this technology has the potential of empowering every human to become a computer programmer with far-range implications. Outreach activities are planned to discuss them (see Section 2.2).

3. QUALITY AND EFFICIENCY OF THE IMPLEMENTATION

3.1 QUALITY AND EFFECTIVENESS OF THE WORK PLAN, ASSESSMENT OF RISKS AND APPROPRIATENESS OF THE EFFORT ASSIGNED TO WORK PACKAGES

The table below lists work packages (WP), the tasks, and dependencies (Dp). I have organized research under this action to start simultaneously with WP1 and WP3 to address at the same time the first steps of the proposed system while working in collaboration with the host group on the pragmatic aspects of the Hanabi environment. The secondment at SFI has been placed at the beginning of the second semester to discuss with experts in Complex Systems any issues that I could face during WP1. WP2 builds on WP1, but I will start working on T2.1 even before the end of WP1 to mitigate the risk associated with the search of the Γ combinator. The automatic learning aspects of WP3 start after there is a learning system capable of solving bandit tasks (M2). Management activities will take place during the entire length of the project, so too dissemination activities connected to each stage.

WP	Task name	Task description	Training Obj.	Dp	Dlvr
WP1 Emergent Evolution	T1.1 Scaling up the simulation	The fellow will add support for GPU computation and include new combinators in CC.	TO1.3 (Complx.)	—	D1.1 D1.2
	T1.2 Exploring the effects of mutations	The fellow will add mutation reactions to Combinatory Chemistry and explore its effects.	TO1.3 (Complx.)	—	
	T1.3 Visualization	The fellow will develop visualization tools to showcase the system.	TO2.3 (Comm.)	—	
WP2 Reservoir Learning	T2.1 Identification of the Γ combinator	The fellow will manually design a Γ combinator to investigate its properties and the likelihood of it emerging autonomously. This task is part of a risk mitigation measure integrated into the work plan due to the high risk and importance of milestone M1.	TO1.3 (Complx.), TO1.4 (Emgt. Learning)	—	D2.1 D2.2
	T2.2 Encoding/decoding	The fellow will design an encoding/decoding scheme for Hanabi's states and actions.	TO1.3 (Compl.), TO1.4 (Emgt. Learning)	T2.1	

	T2.3 Feeding rewards	The fellow will design a method to reinforce the system when it produces desired actions.	TO1.3 (Complx.), TO1.4 (Emgt. Learning)	M1	
WP3 Playing Hanabi	T3.1 Communication in Hanabi	The fellow and the host group will study the pragmatic messages afforded by players' actions in Hanabi.	TO1.1 (Pragma.), TO1.2 (Non-verbal Comm.), TO2.1 (Collab.)	—	D3.1
	T3.2 Learned rules	The fellow will evaluate the learning system in the Hanabi learning environment.	TO1.1 (Pragma.), TO1.2 (Non-verbal Comm.), TO1.4 (Emgt. Learning)	M2	
	T3.3 Interpretation	The fellow and the host group will interpret the rules learned by the system to find whether the system is learning intentions.	TO1.1 (Pragm.), TO1.2 (Non-verbal Comm.), TO2.1 (Collab)	T3.2	
WP4 Secondment	T4.1 Collaborations with SFI	During the secondment at SFI, the fellow will collaborate with experts in Complexity, AI, and Evolutionary Theory.	TO1.3 (Complx.), TO2.1 (Collab.), TO2.3 (Comm.)	—	D4.1
WP5 Management	T5.1 Project management	The fellow and the supervisor will organize and attend project management meetings to monitor progress in the project, write project reports and design the CDP, DEP, and DMP.	TO2.2 (Lead.), TO2.4 (Mngmt.), TO2.5 (Business Awr.)	—	D5.1 D5.2
WP6 Dissemination and Comm.	T6.1 Scientific Communication	The fellow and collaborators will submit conference papers and a multidisciplinary journal article.	TO2.2 (Leader.), TO2.3 (Comm.)	—	D6.1
	T6.2 Public Communication	The fellow will engage the general public in the research outcomes and foster public debate on the project objectives.	TO2.2 (Leader.), TO2.3 (Comm.)	—	

List of major Milestones: **M1**, Emergence of an open-ended evolutionary process (WP1); **M2**, Learning in bandit tasks (WP2); **M3**, 5% win rate at 5P Hanabi (WP3)

List of Deliverables (Dlvr): **D1.1**, Source code of Combinatory Chemistry expanded with mutations and more extensive simulations; **D1.2**, Report on emergent evolution; **D2.1**, Source code of learning system, **D2.2**, Report on learning system; **D3.1**, Report on playing Hanabi; **D4.1**, Report on Secondment; **D5.1**, Data Management Plan (DMP); **D5.2**, Career Development Plan (CDP); **D6.1**, Dissemination and Exploitation Plan (DEP).

Month	01-03	04-06	07-09	10-12	13-15	16-18	19-21	22-24
WP1 Emergent Evolution	T1.1	T1.1	T1.2	T1.2/3				
WP2 Reservoir Learning			T2.1	T2.2	T2.2/3	T2.3		
WP3 Playing Hanabi	T3.1	T3.1			T3.2	T3.2	T3.3	T3.3
WP4 Secondment			T4.1					
WP5 Management	T5.1	T5.1	T5.1	T5.1	T5.1	T5.1	T5.1	T5.1
WP6	Dissemination	T6.2	T6.2	T6.1	T6.1	T6.2	T6.1	T6.2
	<i>Conferences</i>			ALIFE			ACL	
	<i>Journal</i>							PNAS
	<i>Public</i>	Podcast	Escolab			Night	Radio	Media
Deliverables		D5.1	D1.1/2	D4.1	D5.2	D2.1/2	D5.3	D3.1
Milestones			M1		M2		M3	

In the following table, I have identified the most critical **risks** according to their likelihood and potential impact:

Risk	Lik.	Impact	Contingency Plan
After scaling up and adding mutations, we do not observe emergent evolution.	Low	High	I will investigate alternative mutation mechanisms. Also, thanks to my secondment at SFI, I will be able to discuss potential causes with experts in the field.
The evolutionary process does not lead to the hypothesized Γ combinator.	Med	Med	I will investigate whether alternative emergent structures could fulfill a similar function. I will also explore adding the manually engineered combinator in T2.1 to the system.
The RL system cannot learn Hanabi to a satisfactory level.	Med	Med	I will discuss with the host team whether we can simplify the game's dynamics to improve chances of success while still focusing on the phenomena of interest.

The RL system learns to play very well Hanabi without actually learning about intentions.	Med	Med	I will identify loopholes that allow the agents to play without learning intentions. Together with the host team, we will propose ways to improve the setup and publish it for the interest of the AI community.
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3.2 QUALITY AND CAPACITY OF THE HOST INSTITUTIONS AND PARTICIPATING ORGANIZATIONS, INCLUDING HOSTING ARRANGEMENTS

Hosting arrangements: The UPF offers me and my research project the opportunity to integrate into a close-knit interdisciplinary and international environment of researchers with extensive experience in administering individual and collaborative research projects at the national and EU level (“HR Excellence in Research” seal from the EC). I expect a smooth integration with the host team due to earlier collaboration with Prof. Boleda on two high-impact conference proceeding papers and the co-supervision of an MSc. Student (cf. CV). I will meet Prof. Boleda at least twice a month to discuss progress and any issues arising within my project. I will give three research talks, in the beginning, at the middle, and at the end of my project, to gather feedback from the group and bootstrap our collaboration. Further dialogue, meeting, training, and networking opportunities will be provided by the regular scientific events at Prof. Boleda’s group: The regular group meetings and reading group, as well as the group series for internal discussion and presentation trials, the seminar with talks by external researchers, and on-demand collaborative tutorials about project-related methods and techniques. Practical arrangements will be overseen by the UPF administration, with extensive experience in hosting foreign researchers (also see Sections 3.3 and 4): I will have access to an office, all the necessary equipment for the completion of the project, including a high-computing cluster with GPU units, and IT and administrative support. I will be fully covered by the Spanish legislation throughout the fellowship, providing standard social security benefits.

The SFI is a hub for Complexity research that has been designed to foster dialogue and collaboration between multiple disciplines. To ensure a smooth integration, I will meet with Prof. Mitchell on arrival and on a bi-weekly basis afterward to discuss effective ways to advance my project. I will also meet Hilary Scolnik, the visitor program manager, to ensure that I will have access to the institute facilitates and networks, and up-to-date calendar of events.

Quality and capacity of the participating organizations: The infrastructure/facilities offered by the UPF perfectly respond to the needs set out for the execution of the project. Firstly, the Department of Translation and Language Sciences is located in the new UPF Campus de la Comunicació, within the innovation/technological district 22@Barcelona, where I will have my office and all the necessary state-of-the-art hardware (including a GPU-based computing framework) and software for smooth execution of the project. Secondly, I will be granted access to the campus libraries and IT services, UPF cluster with CPU and GPU, events (seminars, workshops, conferences, summer school), language training, and cultural/sports services. I will likewise have remote access to the other UPF libraries, to inter-library lending, IT Services, UPF open access repository (<http://repositori.upf.edu/>), and the university’s Scientific Output Portal (<http://argos.upf.edu/>). Finally, the UPF has a Research Service and an Innovation Office whose staff is expert in the procedures followed and standards adhered to in EU-funded projects.

The SFI’s main Cowan Campus at 1399 Hyde Park Road provides 23,500 square feet of facilities, surrounded by 32 park-like acres. The Santa Fe Institute (SFI) has retained all but a fraction of the site in this natural state as part of its commitment to creating an environment conducive to scientific research. The building includes a large conference room seating up to 75, several small meeting rooms, administrative offices for a staff of 25, shared office space for up to 50 scientists including postdoctoral fellows, computer facilities, including 10 SMP servers, and more than 50 Linux, Windows and Macintosh workstations, plus a 4-node 256-core cluster for large-scale computation, and library space with about 15,000 volumes representing the research profile of SFI’s science, and access to online tools such as First Search, Web of Science, JSTOR, and WorldCat, a database of 50 million books in libraries worldwide. The Institute has an active visitor, workshop, and working group program with 300-400 visitors each year and 25-30 workshop and working group events supported by a professional events management staff.

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